MEHUL RATHORE

Robotics Engineer

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EDUCATION

Indian Institute of Technology Madras — Nov 2022 - Jul 2026

Bachelor of Technology in Mechanical Engineering — *Chennai, India*

COURSEWORK

Linear Algebra, Kinematics & Dynamics of Machinery, Measurements & Controls, AI in Manufacturing, Soft Robotics

EXPERIENCE

Actuarial Analyst — Data and ML — Jul 2026 - Present

ICICI Lombard — *Mumbai, India*

- Built end-to-end **ML pipelines** in Python on large structured datasets (automated feature extraction, distribution analysis, data validation) and audited models for risk modeling using **SHAP, LIME, and PDPs** to assess feature importance and decision boundaries.

Computer Vision Intern — Aug 2024 - Apr 2025

Dorm Doodle — *Chennai, India*

- Processed and analyzed **2,000+ labeled facial images** using **MobileNetV2 embeddings**, **LBP** descriptors, and color histogram features for skin tone classification.
- Generated **1,800+ dimensional feature vectors** by combining handcrafted and deep visual features, enabling robust representation of skin-tone characteristics.
- Fine-tuned an **XGBoost classifier**, achieving **90% accuracy** and a **0.91 F1-score** through feature engineering and model optimization.

PROJECTS

Embedded LiDAR Perception System | Solidworks, C++, Arduino, LiDAR, Python, CloudCompare — June 2026

- Designed mounts and housing in **SolidWorks** for a custom LiDAR prototype built around an **Arduino**, **VL53L0X ToF sensor**.
- Designed a **slip ring** for continuous 360° rotation with uninterrupted power and signal transmission.
- Assembled a **pan–tilt scanner** (two servos on an external 5 V supply) running a raster **3D sweep** with a settle delay before each reading; streams `pan,tilt,dist_mm` over serial at **115200 baud**.
- Tuned the **VL53L0X timing budget** to **20 ms** for faster reads, trading precision for speed, and applied a distance-offset calibration.
- Developed a **Python** pipeline: serial logging to CSV, invalid-reading filtering, **spherical-to-Cartesian** conversion, and **Open3D** `.ply` export (∼1.6k points), inspected in **CloudCompare**.

Automated Paint Defect Detection and Correction | ROS2, RViz, TF2, Gazebo, OpenCV, ResNet18 — Feb 2025

- Architected a ROS2 robotic inspection pipeline on a simulated **UR5e** manipulator in Gazebo, integrating camera perception, CNN inference, and motion planning into an automated defect detection workflow.
- Built a custom **ROS2** Python inference node using PyTorch ResNet18 to classify surface defects from live camera topics and publish binary predictions for downstream robotic actions.
- Configured **ros2_control** with the Gazebo hardware interface and authored a custom Xacro robot description mounting a wrist-mounted camera, enabling synchronized sensor-robot co-simulation in Gazebo Harmonic.
- Integrated MoveIt2 motion planning with a **fine-tuned** ResNet18 classifier (**91% accuracy**, **0.85 F1-score** on DAGM 2007), executing autonomous pick-and-place sequences upon defect detection via MoveItPy.

E-Yantra Robotics Competition | Python, OpenCV, CoppeliaSim — Sep 2026 - Present

- Developed a **lane detection pipeline in Python/OpenCV** (HSV/HLS colour thresholding, Hough lines, bird's-eye perspective warp with fallback trapezoid), achieving **96.9% ego-lane accuracy** and **23.5 px mean lane-centre error** on 3,708 scored frames at **285 FPS**.
- Built a **lane-identification module** that determines the ego lane by classifying solid vs dashed boundary lines on either side of the vehicle.
- Tuned a **Stanley controller** in **CoppeliaSim** (cross-track gain, softening term) for lateral tracking; on a **0.2 m lane-change step** it settled with ∼**2.5% overshoot** and **near-zero steady-state error**.

5DOF Inspection Light Holder Mechanism | Solidworks **Jul 2026**

- Designed a **5-DOF light positioning bracket** for a **machine vision inspection system** in **SolidWorks**; sized a **2-link revolute arm** (L1=L2=45mm) via **workspace reachability analysis** inside a **60×120×300mm** envelope, and verified **5-DOF mobility** with **Gruebler's formula** (6 links, 5 joints: 1 prismatic + 4 revolute).
- Engineered **fastener-free friction-arrest prismatic and revolute joints** that allow **tool-free manual adjustment** with **rigid position locking**, cutting part count and setup effort; validated the **multi-body assembly** with constrained mates and motion limits.

Robotics Motion Toolkit | C++, Eigen, OOP, Robotics Kinematics, Motion Planning **May 2026**

- **Implemented** Forward Kinematics using Denavit–Hartenberg (DH) parameters and Product of Exponentials (PoE), validating end-effector pose estimation across both formulations.
- **Developed** a $6 \times n$ Geometric Jacobian and Damped Least Squares (DLS) inverse kinematics solver for robust manipulator control under near-singular configurations.

EXTRACURRICULAR

Aeroclub **Aug 2024 - Dec 2024**

Project Member *Centre for Innovation, IIT Madras*

- Analyzed the existing **VTOL flight controller codebase** and implemented firmware modifications to support evolving project requirements.
- Integrated **IMU and position sensors** with the flight controller, verifying sensor communication and enabling reliable state estimation during prototype testing.

TECHNICAL SKILLS

- **Robotics:** ROS2, TF2, State Estimation, Sensor Fusion, PID Controller, Arduino, Hardware
- **Simulation / Perception:** Gazebo, RViz2, OpenCV, Fusion 360, Solidworks, CoppeliaSim, CloudCompare
- **Machine Learning:** PyTorch, CNNs, VAE, LSTM, PySpark, PDP, SHAP
- **Languages:** Python, C, C++, R
- **Tools:** Linux, Git, MATLAB, VS Code, Databricks, Excel